Homework of AI and Machine Learning 2026

1. Given a dataset in the file named clsn_trn.csv for training, please train a maximum likelihood estimator to learn a classifier for the data with 7 classes. Test your classifier on the testing dataset in the file named clsn_tst.csv. Please (1) show the discriminant functions $g_i(\boldsymbol{x})$, $i = 1\ldots7$. (2) Please calculate the determinant of covariance matrix in $g_i(\boldsymbol{x})$, $i = 1\ldots7$. In other words, please calculate $|\Sigma_i| \equiv \det(\Sigma_i)$, $i = 1\ldots7$. (3) generate a file named class1_result.csv to contain the testing results based on the testing data in the file clsn_tst.csv. Each row of the data in class1_result.csv should contain [testing data x], [class]. (The answer is in the file named clsn_ans.csv. Please compare your answers class1_result.csv with this file.)